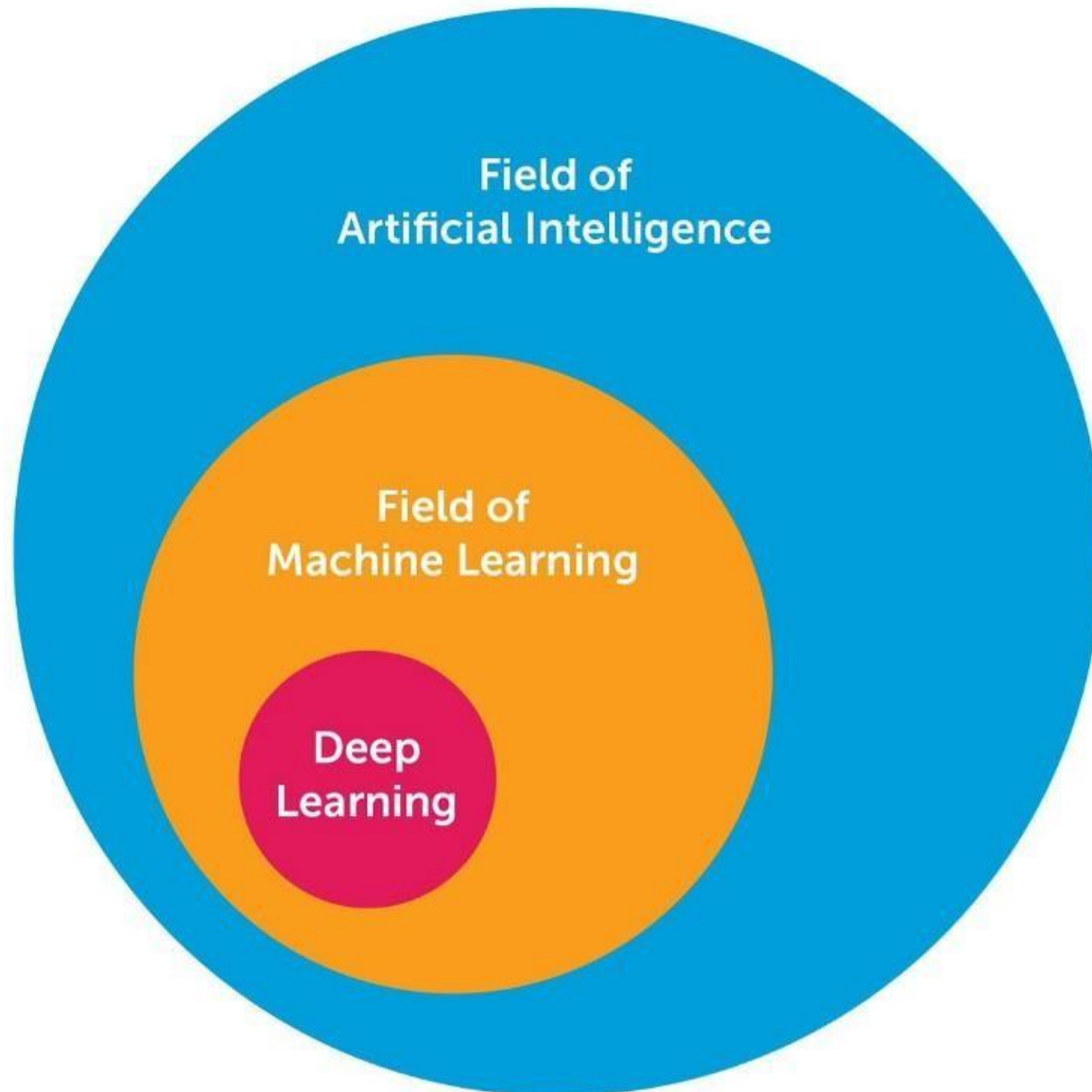


Machine Learning and its Applications

Introduction

- AI, ML and DL



Introduction

- **AI, ML and DL :** Artificial Intelligence (AI) is the branch of computer science, which is used to create intelligent machines. Artificial intelligence refers to human intelligence or mimics human behavior by machines.
- Machine learning (ML) is referred to as a subset of Artificial intelligence (AI). It allows a computer to handle situations via training, analysis, observation, and experience. All the machine learning counts as Artificial intelligence, but all AI does not count as machine learning. It is considered one of the best tools of artificial intelligence that is suitable for business.
- Deep learning (DL) is referred to as the subset of machine learning. It is generally referred to as a deep artificial neural network, and these are the algorithm sets that are extremely accurate for problems like sound recognition, image recognition, etc. Deep learning is also defined as it enables a computer to learn without being programmed to do so.

Introduction

- What is Machine Learning (ML)?
 - Machine learning is a subfield of artificial intelligence, which is broadly defined as the capability of a machine to imitate intelligent human behavior. Artificial intelligence systems are used to perform complex tasks in a way that is similar to how humans solve problems.
 - Machine learning (ML) is a branch of artificial intelligence (AI) and computer science that focuses on the using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
 - Tom Mitchell provides a more modern definition: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ." Example : I will make a simple example to understand better .

Introduction

➤ What is Machine Learning (ML)?

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Introduction

- Flavors of Machine Learning

Introduction

- Flavors of Machine Learning
 - Supervised Learning
 - Unsupervised Learning
 - Semi Supervised Learning
 - Reinforcement Learning

Introduction

- Have you already been benefitted by ML?
 - Google search and ranked pages
 - Spam E-mail
 - Recommendations on Amazon, Flipkart, Movielens
 - Related advertisements on websites

Applications

- Natural Language Processing
 - Autonomous Tagging of Stackoverflow Questions
 - Make a multi-label classification system that automatically assigns tags for questions posted on a forum such as StackOverflow or Quora.
 - Dataset: StackLite or 10% sample

Applications

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 - The purpose of this project is to implement and train machine learning algorithms to automatically assess and grade essay responses.
 - Dataset: Essays with human graded scores

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 - The purpose of this project is to implement and train machine learning algorithms to automatically assess and grade essay responses.
 - Dataset: Essays with human graded scores
 - Sentence to Sentence semantic similarity
 - Can you identify question pairs that have the same intent or meaning?
 - Dataset: Quora question pairs with similar questions marked

Applications

- Natural Language Processing
 - Fight online abuse
 - Can you confidently and accurately tell whether a particular comment is abusive?
 - Dataset: Toxic comments on Kaggle

Applications

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 - Fight online abuse
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 - Open Domain question answering
 - Can you build a bot which answers questions according to the student's age or her curriculum?
 - Facebook's FAIR is built in a similar way for Wikipedia.
 - Dataset: NCERT books for K-12/school students in India, NarrativeQA by Google DeepMind and SQuAD by Stanford

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 - Dataset: NCERT books for K-12/school students in India, NarrativeQA by Google DeepMind and SQuAD by Stanford
 - Social Chat/Conversational Bots
 - Can you build a bot which talks to you just like people talk on social networking sites?
 - Reference: Chat-bot architecture
 - Dataset: Reddit Dataset

Applications

- Natural Language Processing
 - Copy-cat Bot
 - Generate plausible new text which looks like some other text
 - Obama Speeches? For instance, you can create a bot which writes some new speeches in Obama's style
 - Trump Bot? Or a Twitter bot which mimics @realDonaldTrump
 - Narendra Modi bot saying "doston"? Start by scrapping off his Hindi speeches from his personal website
 - Example Dataset: English Transcript of Modi speeches

Applications

- Applications that can't be programmed by hand
 - Handwriting Recognition

Applications

- Applications that can't be programmed by hand
 - Natural Language Processing
 - Machine Translation

Applications

- Applications that can't be programmed by hand
 - Handwriting Recognition
 - Natural Language Processing
 - Machine Translation
 - Word Sense Disambiguation
 - "I am taking aspirin for my cold" the disease sense is intended
 - "Let's go inside, I'm cold" the temperature sensation sense is meant
 - "It's cold today, only 2 degrees", implies the environmental condition sense.

Applications

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 - Part-of-Speech Tagging
 - "And now for something completely different"
 - [('And', 'CC'), ('now', 'RB'), ('for', 'IN'), ('something', 'NN'), ('completely', 'RB'), ('different', 'JJ')]
 - Here we see that and is CC, a coordinating conjunction; now and completely are RB, or adverbs; for is IN, a preposition; something is NN, a noun; and different is JJ, an adjective.

Applications

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 - Sentiment Analysis

Applications

- Applications that can't be programmed by hand
 - Natural Language Processing
 - Named-Entity Recognition
 - Teddy Roosevelt was a great president
 - Teddy bears are on sale

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 - Speech Recognition, etc.
 - Speech recognition is the ability of a machine or program to identify words and phrases in spoken language and convert them to a machine-readable format.
 - The most frequent applications of speech recognition include voice dialling and voice search.

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 - The most frequent applications of speech recognition include voice dialling and voice search.
 - Text Summarization
 - Extractive Summarization
 - Abstractive Summarization

Celebrities [2]



Celebrities [2]



Pose Guided Person Image Generation [3]



Ground truth



Generated

Pose Guided Person Image Generation [3]



(a) DeepFashion

(b) Market-1501



(c) Generating from a sequence of poses

CycleGAN [4]

Zebras $\leftrightarrow$ Horses



zebra $\rightarrow$ horse



horse $\rightarrow$ zebra

Text to Image [5]

This flower has long thin yellow petals and a lot of yellow anthers in the center

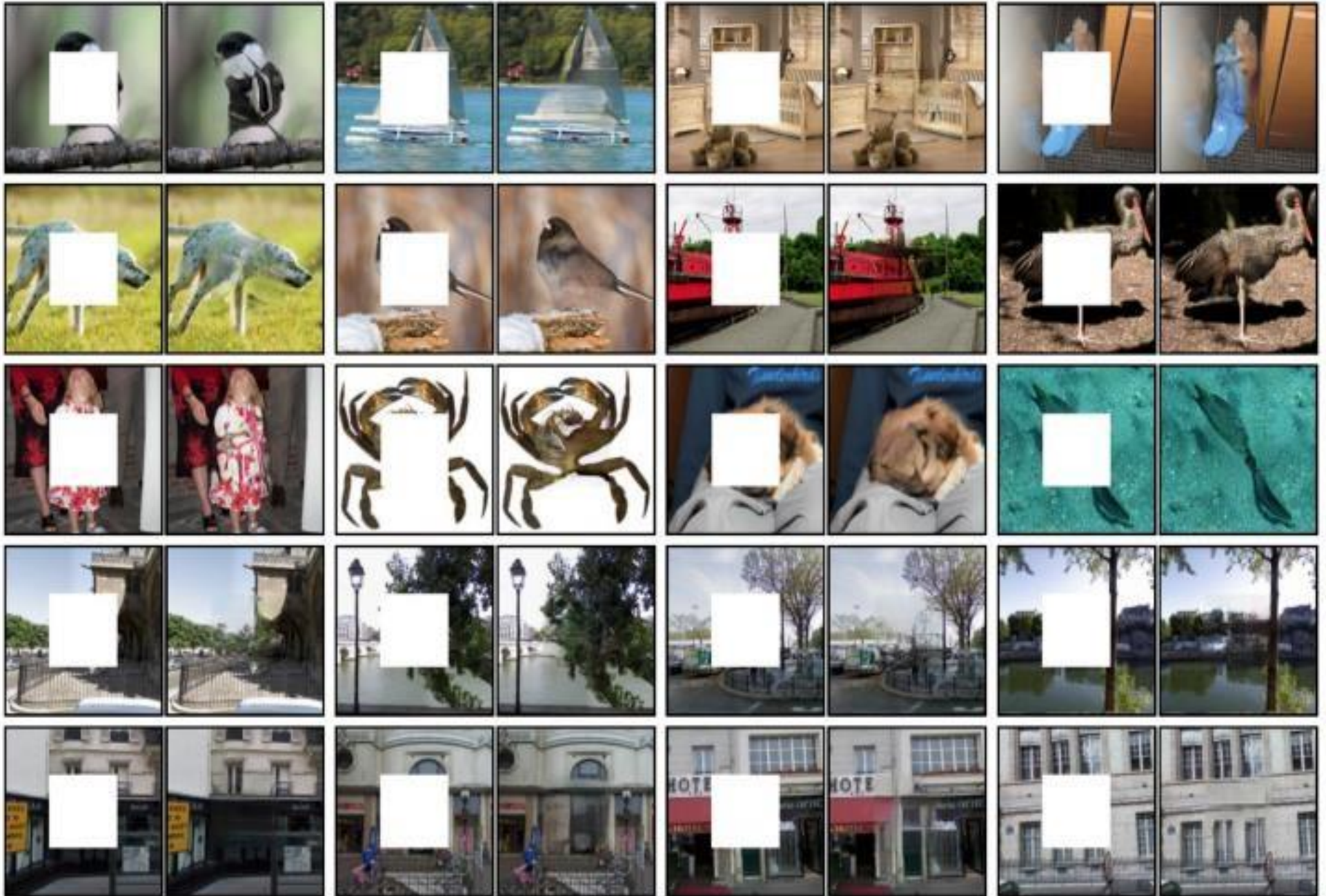
Stage-I



Stage-II



Image Inpainting [6]



DiscoGAN [7]



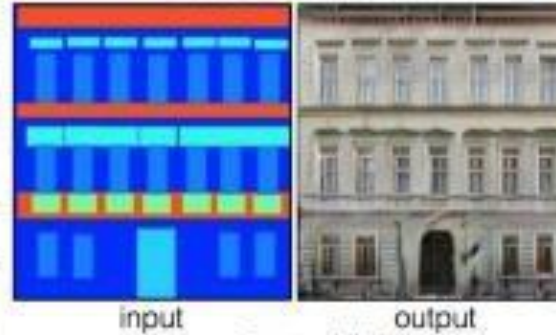
(b) Handbag images (input) & **Generated** shoe images (output)

Pix2Pix [8]

Labels to Street Scene



Labels to Facade



BW to Color



Aerial to Map



Day to Night

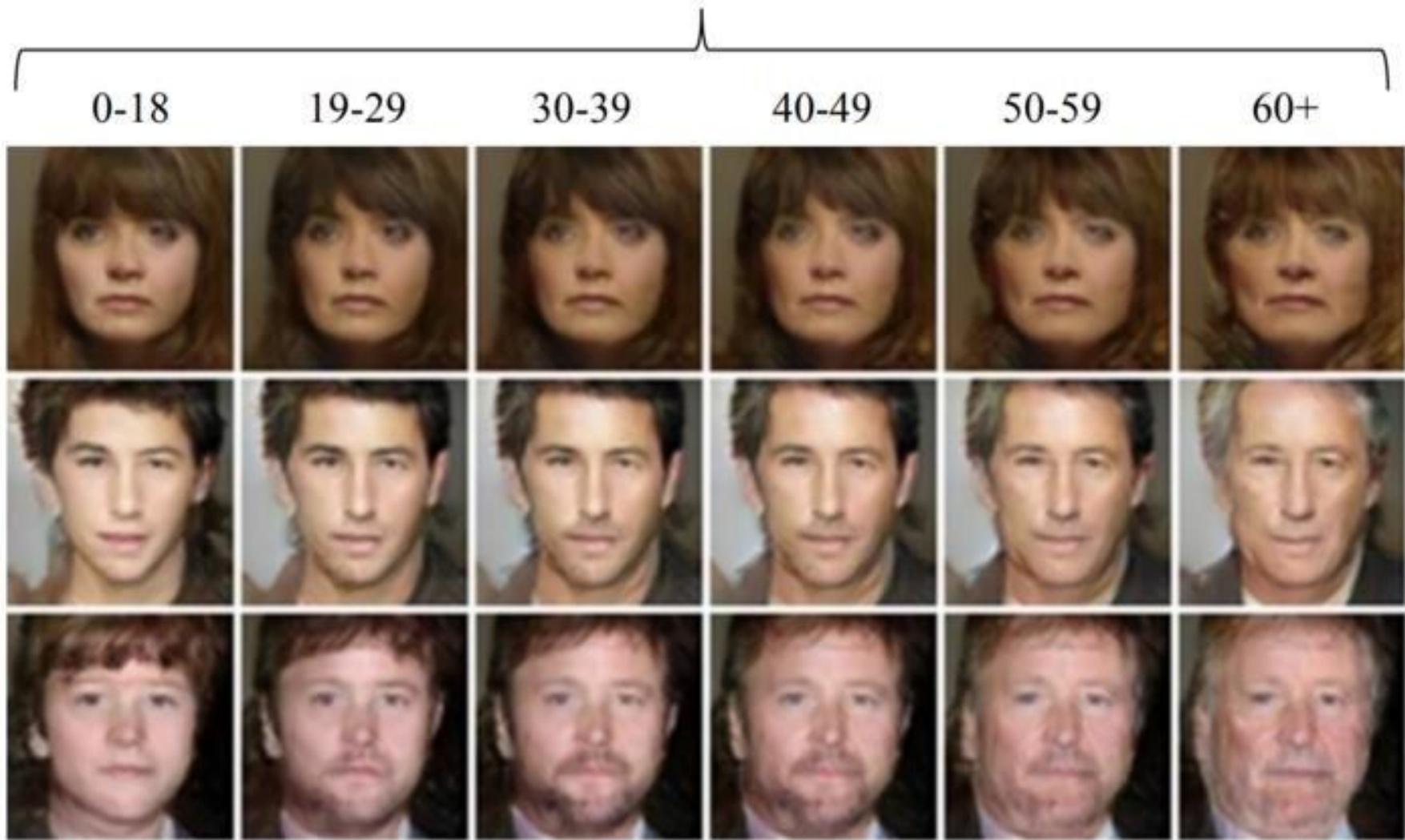


Edges to Photo



Face aging (Age-cGAN) [9]

Face Aging



Unsupervised Domain Adaptation by Backpropagation [10, 11]

- Learning a discriminative classifier or other predictor in the presence of a shift between training and test distributions is known as domain adaptation (DA).



Figure 2. Examples of domain pairs used in the experiments. See Section 4.1 for details.

Applications

- Applications that can't be programmed by hand
 - Computer Vision
 - Security and Surveillance

Applications

- Applications that can't be programmed by hand
 - Computer Vision
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 - Age Invariant Face Recognition

Applications

- Applications that can't be programmed by hand
 - Computer Vision
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 - Age Invariant Face Recognition
 - Gesture Analysis

Applications

- Applications that can't be programmed by hand
 - Computer Vision
 - Security and Surveillance
 - Age Invariant Face Recognition
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 - Autonomous Vehicle

Applications

- Applications that can't be programmed by hand
 - Computer Vision
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 - Gender Recognition

Applications

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Applications

- Applications that can't be programmed by hand
 - Computer Vision
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 - Gender Recognition
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 - Age Prediction

Applications

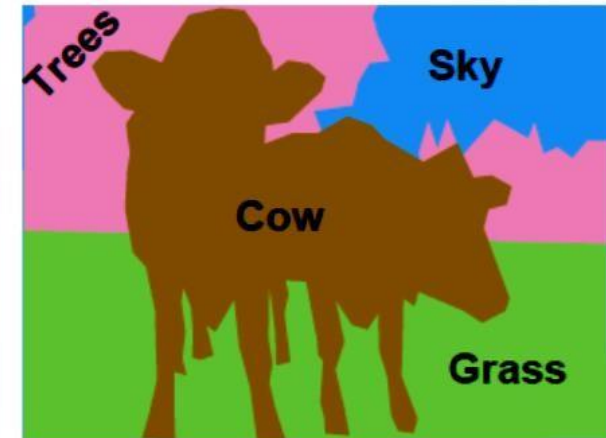
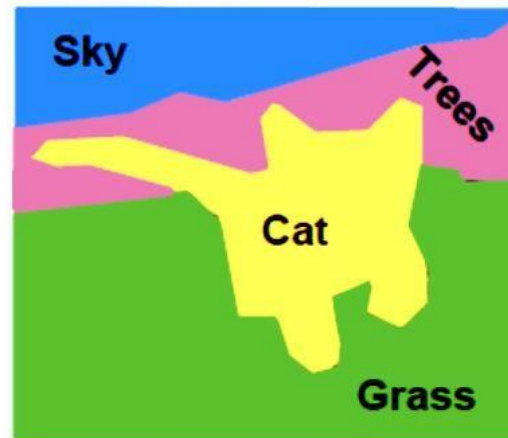
- Applications that can't be programmed by hand
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 - Age Prediction
 - Image Classification (ImageNet Large Scale Visual Recognition Competition (ILSVRC))

Semantic Segmentation

Semantic Segmentation

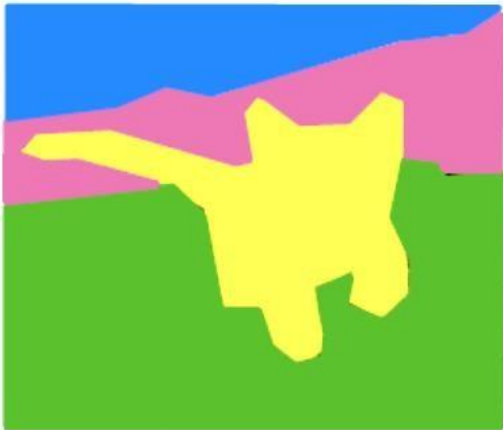
Label each pixel in the image with a category label

Don't differentiate instances, only care about pixels



Instance Segmentation

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

**Classification
+ Localization**



CAT

Single Object

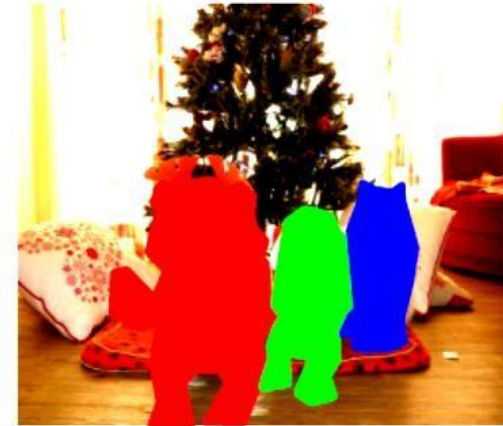
**Object
Detection**



DOG, DOG, CAT

Multiple Object

**Instance
Segmentation**



DOG, DOG, CAT

[This image is CC0 public domain](#)

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 - Image and Video Captioning

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 - Video Summarization

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 - CBIR (three variants)

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 - Instance Segmentation
 - Image and Video Captioning
 - Video Summarization
 - CBIR (three variants)
 - Driver's Drowsiness Detection

Applications

- Applications that can't be programmed by hand
 - Computer Vision
 - Emojinator (Emoji Classifier)

Applications

- Personalized Websites

Applications

- Personalized Websites
- Medical Domain
 - Lung Cancer Detection (Nodules from CT Scan Images)
 - Bone X-ray

Applications

- Personalized Websites
- Medical Domain
- Finance

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System
- Optimized Routing in IP Networks

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System
- Optimized Routing in IP Networks
- Wireless Sensor Network

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System
- Optimized Routing in IP Networks
- Wireless Sensor Network
- Friend Recommendation

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System
- Optimized Routing in IP Networks
- Wireless Sensor Network
- Friend Recommendation
- Tag Recommendation

Applications

- Personalized Websites
- Medical Domain
- Finance
- Intrusion Detection System
- Optimized Routing in IP Networks
- Wireless Sensor Network
- Friend Recommendation
- Tag Recommendation
- Solar Intensity Prediction

Applications

- Personalized Websites
- Medical Domain
- Finance
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- Friend Recommendation
- Tag Recommendation
- Solar Intensity Prediction
- Weather Prediction

Applications

- Personalized Websites
- Medical Domain
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- Intrusion Detection System
- Optimized Routing in IP Networks
- Wireless Sensor Network
- Friend Recommendation
- Tag Recommendation
- Solar Intensity Prediction
- Weather Prediction
- Soil Moisture Estimation

Applications

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- Medical Domain
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- Solar Intensity Prediction
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- Soil Moisture Estimation
- Oil Spill Detection

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- Land use Land Cover Classification

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- Land use Land Cover Classification
- Cyclone Path Prediction

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- Robotics

Applications

- Music Recommender System

Applications

- Music Recommender System
- LipNet

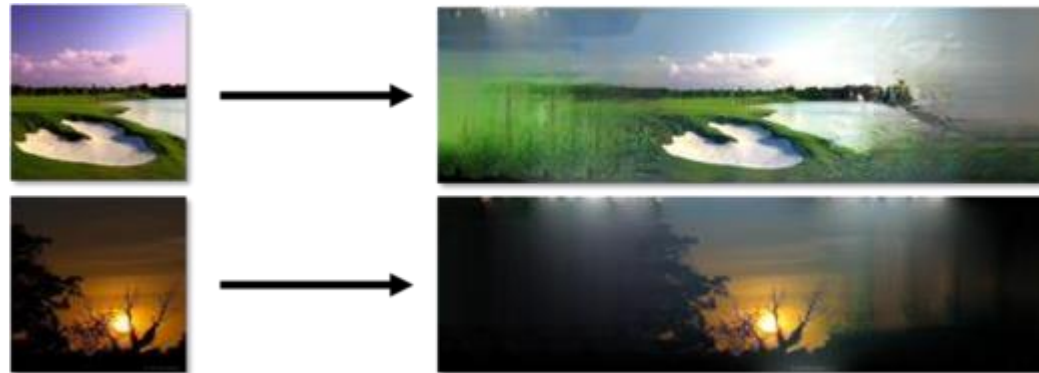
Applications

- Music Recommender System
- LipNet
- Image Enlarging



Applications

- Music Recommender System
- LipNet
- Image Enlarging
- Image Outpainting
 - Imagine you have a half image of a scene and you wanted the full scenery, well that's what image outpainting can do



Applications

- Music Recommender System
- LipNet
- Image Enlarging
- Image Outpainting
- Audio Visual Scene Aware Dialog (DSTC)
 - The goal is to generate a complete and natural response to a question about a scene, given video and audio of the scene and the history of previous turns in the dialog.
 - To answer successfully, agents must ground concepts from the question in the video while leveraging contextual cues from the dialog history.

References

1. <https://towardsdatascience.com/the-10-deep-learning-methods-ai-practitioners-need-to-apply-885259f402c1>
2. Progressive Growing of GANs for Improved Quality, Stability, and Variation
3. Ma, Liqian, et al. "Pose guided person image generation." Advances in Neural Information Processing Systems. 2017.
4. <https://github.com/junyanz/CycleGAN>
5. <https://github.com/hanzhanggit/StackGAN>
6. <https://github.com/pathak22/context-encoder>
7. <https://github.com/carpedm20/DiscoGAN-pytorch>
8. <https://github.com/phillipi/pix2pix>
9. Antipov, Grigory, Moez Baccouche, and Jean-Luc Dugelay. "Face aging with conditional generative adversarial networks." Image Processing (ICIP), 2017 IEEE International Conference on. IEEE, 2017.
10. <https://pdfs.semanticscholar.org/presentation/f4af/de757b9dfc697d149e95cb193aa4749530e2.pdf>
11. Ganin, Yaroslav, and Victor Lempitsky. "Unsupervised domain adaptation by backpropagation." arXiv preprint arXiv:1409.7495 (2014).

Disclaimer

- Content of this presentation is not original and it has been prepared from various sources for teaching purpose.